Artificial Intelligence (AI)

LAB# 3

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CMS: 023-23-0115

Exercise 1:

Rules:

```
sum (First, Last, 0) :-
    First > Last.
sum (First, Last, First) :-
    First =:= Last.
```

```
sum (First, Last, Sum) :-
        First < Last,
        Temp is First + 1,
        sum (Temp, Last, S),
        Sum is First + S.
```

Through Input:

Rule:

```
sum_input :-
    write ('Enter the first number: '), read (First),
    write ('Enter the Last number: '), read (Last),
    sum (First, Last, Sum),
    write ('Sum of numbers from: '),
    write (First), write ('  to  '), write(Last), write('  is: '),
    write (Sum).
```

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```
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sum_input :-
    write('Enter the first number: '),
    read(First),
    write('Enter the Last number: '),
    read(Last),
    sum(First, Last, Sum),
    write('Sum of numbers from: '),
    write(First), write('  to  '), write(Last), write('  is: '),
    write(Sum).
```

```
sum(0, 4, N)
N = 10
false

sum_input.
Enter the first number:
1
Enter the Last number:
3
Sum of numbers from: 1 to 3 is: 6
true
false

?- sum_input.
```

Exercise 2:

Facts:

```
man(ahmed).
man(ali).
man(umar).
man(faizan).
man(hassan).
man(kashif).
woman(aisha).
woman(sadia).
woman(nida).
woman(komal).
woman(kinza).
woman(zara).
woman(sara).
```

```
parent(ahmed, ali).
parent(aisha, ali).
parent(ahmed, zara).
parent(aisha, zara).
parent(ahmed, umar).
parent(sadia, umar).
parent(ali, faizan).
parent(nida, faizan).
parent(ali, hassan).
parent(nida, hassan).
parent(umar, kashif).
parent(nida, kashif).
```

```
married(ahmed, aisha).
married(ali, nida).
married(faizan, komal).

married(ahmed, sadia).
married(umar, nida).
married(umar, kinza).
```

Rules:

```
father(F, C):-
    man(F),
    parent(F,C).
mother(M, C):-
    woman(M),
    parent(M, C).
brother(B,X):-
    man(B),
    siblings(B, X).
sister(S, X):-
    woman(S),
    siblings(S, X).
siblings(A, B):-
    parent(P, A),
    parent(P, B),
    A \= B.
```

```
cousin_brother(C, X):-
    man(C),
    parent(P1, C),
    parent(P2, X),
    siblings(P1, P2).
cousin_sister(C, X):-
    woman(C),
    parent(P1, C),
    parent(P2, X),
    siblings(P1, P2).
grandparent(G, C):-
    parent(G, X),
    parent(X, C).
```

```
step_mother(S, C):-
    woman(S),
    married(F, S),
    father(F, C),
    \+ mother(S, C).
step_sister(SS, C):-
    woman(SS),
    step_mother(M, C),
    parent(M, SS),
    SS \= C.
step_brother(SB, C):-
    man(SB),
    step_mother(M, C),
    parent(M, SB),
    SB \= C.
```

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```
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man(ahmed).
man(ali).
man(umar).
man(faizan).
man(hassan).
man(kashif).

woman(aisha).
woman(sadia).
woman(nida).
woman(komal).
woman(kinza).
woman(zara).
woman(sara).

parent(ahmed, ali).
parent(aisha, ali).
parent(ahmed, zara).
parent(aisha, zara).
parent(ahmed, umar).
```

```
father(ahmed, C).
C = ali
C = umar

mother(aisha, C).
C = ali

mother(nida, C).
C = faizan
C = hassan

grandparent(ahmed,_).
true
true
false

grandparent(ahmed,X).
X = faizan
X = hassan
false

sister(zara, X)
X = ali

?- step_mother(sadia, X).
```

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```
parent(ali, faizan).
parent(nida, faizan).
parent(ali, hassan).
parent(nida, hassan).
parent(umar, kashif).
parent(nida, kashif).
parent(umar, sara).
parent(kinza, sara).

married(ahmed, aisha).
married(ali, nida).
married(faizan, komal).

married(ahmed, sadia).
married(umar, nida).
married(umar, kinza).

father(F, C):-
    man(F),
    parent(F,C).
mother(M, C):-
    woman(M),
    parent(M, C).
```

sister(zara, X)

X = ali
false

brother(ali, X)

X = umar
X = zara

cousin_brother(kashif, X)

X = faizan
X = hassan
false

cousin_sister(sara, X)

X = faizan
X = hassan
false

siblings(S, sara)

S = kashif
false

step_mother(kinza, kashif).

true

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```
brother(B,X):-
    man(B),
    siblings(B, X).
sister(S, X):-
    woman(S),
    siblings(S, X).
siblings(A, B):-
    parent(P, A),
    parent(P, B),
    A \= B.
cousin_brother(C, X):-
    man(C),
    parent(P1, C),
    parent(P2, X),
    siblings(P1, P2).
cousin_sister(C, X):-
    woman(C),
    parent(P1, C),
    parent(P2, X),
    siblings(P1, P2).
grandparent(G, C):-
    parent(G, X),
    parent(X, C).
step_mother(S, C):-
```

step_mother(kinza, kashif).

true

step_mother(kinza, X).

X = kashif
false

step_brother(kashif, X).

X = sara
false

step_sister(sara, X).

X = kashif
false

step_sister(zara, X).

X = umar
false

step_mother(aisha, X).

X = umar

step_mother(sadia, X).

X = ali
X = zara
false

```
step_mother(S, C):-
    woman(S),
    married(F, S),
    father(F, C),
    \+ mother(S, C).
step_sister(SS, C):-
    woman(SS),
    step_mother(M, C),
    parent(M, SS),
    SS \= C.
step_brother(SB, C):-
    man(SB),
    step_mother(M, C),
    parent(M, SB),
    SB \= C.
```

step_mother(kinza, X).

X = kashif
false

step_brother(kashif, X).

X = sara
false

step_sister(sara, X).

X = kashif
false

step_sister(zara, X).

X = umar

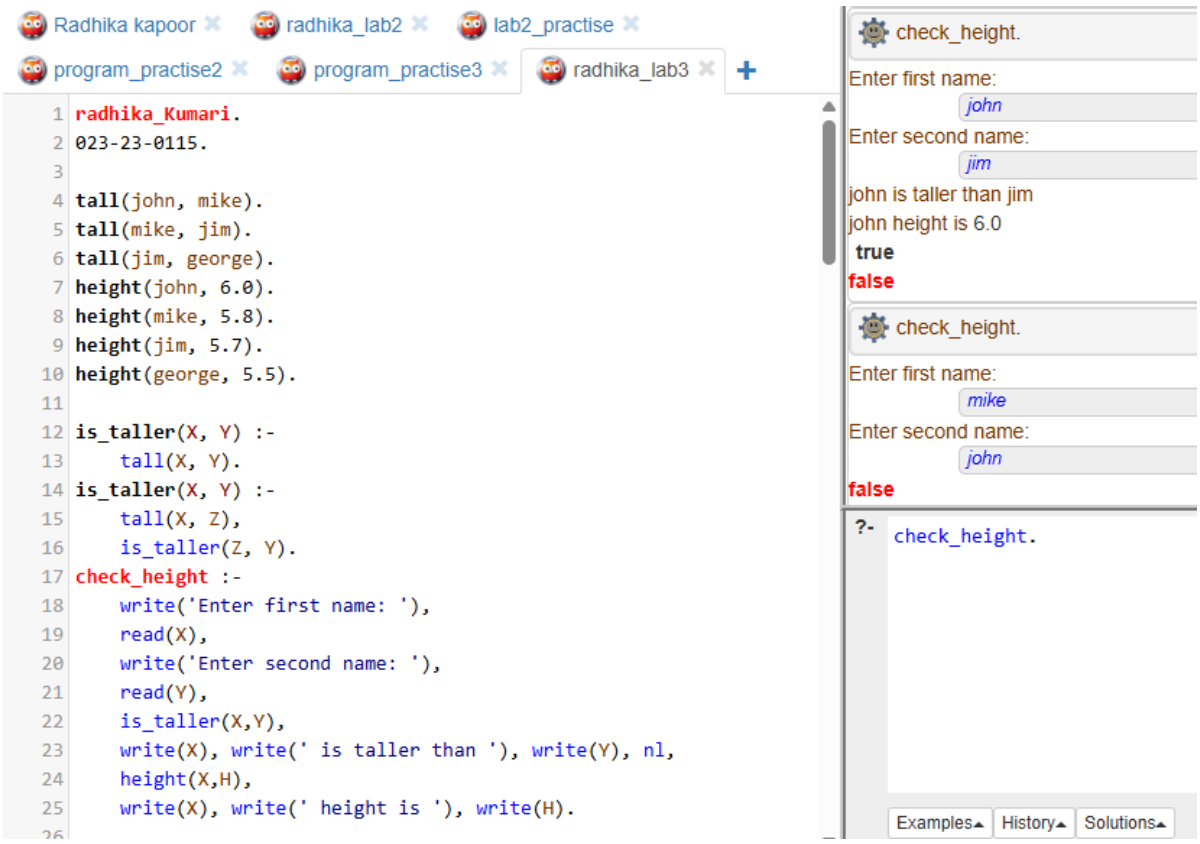
Exercise 3:

Facts:

```
tall(john, mike).
tall(mike, jim).
tall(jim, george).
height(john, 6.0).
height(mike, 5.8).
height(jim, 5.7).
height(george, 5.5).
```

```
is_taller(X, Y) :-
    tall(X, Y).
is_taller(X, Y) :-
    tall(X, Z),
    is_taller(Z, Y).
```

```
check_height :-
    write('Enter first name: '), read(X),
    write('Enter second name: '), read(Y),
    is_taller(X,Y),
    write(X), write(' is taller than '),
    write(Y), nl, height(X,H),
    write(X), write(' height is '), write(H).
```



Exercise 4:

Facts:

```
road(town1,town2).
road(town2,town3).
road(town3,town4).
road(town4,town5).
road(town5,town6).
```

Rules:

```
can_get(T1, T2) :-
    road(T1, T2).
can_get(T1, T2) :-
    road(T1, T3),
    can_get(T3, T2).
```

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road(town1,town2).
road(town2,town3).
road(town3,town4).
road(town4,town5).
road(town5,town6).

can_get(T1, T2) :-
    road(T1, T2).
can_get(T1, T2) :-
    road(T1, T3),
    can_get(T3, T2).
```

```
Enter second name:
    george
jim is taller than george
jim height is 5.7
true
false
can_get(town1, town6).
true
false
can_get(town4, town1).
false
?- can_get(town4, town1).
```

Through input:

Rule:

```
travel :-
    write ('Enter starting town: '),
    read(X),
    write ('Enter destination town: '),
    read(Y),
    ( can_get(X,Y) ->
        write ('yes you can.')
    ;
        write ('you cannot go reverse.')
    ).
```

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travel :-
    write('Enter starting town: '),
    read(X),
    write('Enter destination town: '),
    read(Y),
    ( can_get(X,Y) ->
        write('yes you can.')
    ;
        write('you can not go reverse.')
    ).
```

```
travel.
Enter starting town:
    town3
Enter destination town:
    town6
yes you can.
true
travel.
Enter starting town:
    town6
Enter destination town:
    town2
you can not go reverse.
true
?- travel.
```